Market Equilibrium Analysis Considering Electric Vehicle Aggregators and Wind Power Producers Without Storage Capabilities

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Abstract

This document is devoted to analyze market equilibrium solutions when new market agents, such as renewable/intermittent producers, elastic demands and electric vehicles, are exposed to time-varying locational marginal prices in the context of a competitive electricity market. We consider three main features for these market agents:

1. Load demands are deemed to be elastic, being fully responsive to price changes
2. Renewable-based producers, such as wind farms are not dispatchable in real power and deprived of storage capabilities, and
3. Electric-vehicle retailers -defined as EV aggregators- are able to manage charge-discharge pattern of EV's batteries in order to enhance daily profits due to energy traded in the wholesale electricity market.

After a brief discussion about different economic equilibrium models in power systems (Perfect competition, Nash-Cournot, Stackelberg and Monopoly), two economic equilibrium models are studied in detail. First, we analyze the *perfect competition solution* driven by a benevolent planner in which real and reactive power dispatches as well as the battery charge-discharge schedule aims to maximize the global social welfare. Secondly, we also address the *monopolistic solution* where the total profit of EV aggregators and renewable generators are maximized considering that both producers belong to the same firm.
Solutions were obtained using a multi-period alternate current optimal power flow (AC-OPF) formulation in which nodal reactive power injections can be modified by renewable generators and EV aggregators in order to shift the locational marginal prices to their own benefit. The 24-hour AC-OPF was stated as the maximization problem subject to active and reactive power equilibrium equations as well as battery capacity constraints.
The perfect competition and monopolistic system models were applied to an illustrative 3-node test system. Optimization problems were solved using Matlab's fmincon optimization tool. Results show
Keywords: Optimal Power Flow, Social Welfare, Optimization, Monopolistic, Nash Equilibrium, Stackelberg Game, Locational Marginal Prices, Wind Power, EV Aggregator.

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1. Introduction

In the scope of Smart Grid paradigm [**Include a Reference Here**], we can identify that ongoing power system modernization process is carried out applying four main drivers: 1) energy matrix transformation by allocating large amounts of renewable energy, 2) promoting electric mobility in order to substitute combustion engines in the transportation sector, 3) widespread deployment of smart meters in order to apply demand-response programs, and 4) the deployment of Advanced Metering Infrastructure (AMI) enables the application of real-time pricing schemes to market agents due to their impact on energy losses and congestion.

Renewable energies are being progressively introduced the electricity market in the last few years boost by national police-making energy diversification goals and increasing environmental concerns [Herig, 2003]. Several markets around the world have included renewable resources in their portfolio in spite of the increase in variability and therefore the undesirable effects on system reliability. Thus, since intermittent sources produce energy proportionally with wind speed and solar irradiance, which cannot be turned off or adjusted, they must be regarded as non-dispatchable in real power. As a result, intermittent renewable generation is capable to provide reliable energy to the market since forecasting models have reached certain maturity but with capacity limitations with respect to centralized generation. To solve this drawback, energy storage devices such as batteries can be included in renewable plants. As this solution implies an increased capital expenditure (CAPEX), renewable energy promoters are installing generation facilities without storage capabilities [Root et al., 2017].

Generalized electric mobility have important energy storage capabilities that intermittent sources do not have. Thus, electric vehicle (EV) aggregators can provide capacity and energy as virtual power producers by exploiting business opportunities trading part of the stored energy in the plug-in vehicles in the wholesale electricity market [**Include a Reference Here**]. Under a 24-hour cycle, EV aggregators can decide when and how to charge and discharge the batteries in order to maximize their own profit and transferring part of these benefits to the EV owners. Added to this, EV aggregators and renewable producers can manage reactive power via power electronics or other reactive power compensation devices.

Smart metering can enable sending energy prices to elastic consumers under a real time-basis. In this case, retail aggregators can seek savings by managing demand profiles taking

into account price changes. Real-time prices can be sent to users using locational information accounting not only the cost of electricity in the wholesale markets but also the effects upon losses and line congestion [**Include a Reference Here: Schweppe, Kirschen...**].

Under advanced smart grid paradigm with telemetry based upon Phasor Measurement Units (PMU), Distribution Management System applications such as State Estimation and Optimal Power Flow are capable to compute Locational Marginal Prices (LMP) and send them to market agents using the Advanced Metering Infrastructure (AMI) [Li et al., 2010] [**Include a Reference Here: Heydt Next generation...**].

These new features will affect the structure of electricity markets. Traditionally, if the cost structure of a electricity market is known the market equilibrium points (perfect competition, oligopolistic or monopolistic solution) can be emulated using the Optimal Power Flow tool (either DC or AC formulation) where the marginal prices are acquired by computing the corresponding Lagrange multipliers. There are several approaches to analyze electricity markets from the OPF perspective: Security Economic Dispatch, Security Constrained OPF, Hydrothermal coordination, etc [**Include a Reference Here: Zhu OPF book...**]. The research question raised here is how to evaluate market equilibria (perfect competition and monopolistic solutions) when above mentioned market agents with and without storage capabilities are included in the formulation of the complete AC optimal power flow.

After a brief discussion about different economic equilibrium models in power systems (Perfect competition, Nash-Cournot, Stackelberg and Monopoly), two economic equilibrium models are studied in detail. First, we analyze the *perfect competition solution* driven by a benevolent planner in which real and reactive power dispatches as well as the battery charge-discharge schedule aims to maximize the global social welfare. Secondly, we also address the *monopolistic solution* where the total profit of EV aggregators and renewable generators are maximized considering that both producers belong to the same firm. Both market equilibrium approaches are analyzed and compared with a minimum loss solution dispatch performed by the operator. The 24-hour AC-OPF was stated as the maximization problem subject to active and reactive power equilibrium equations as well as battery capacity constraints.

The paper is structured as follows: **1)** Introduction, **2)** Theoretical Framework, **4)** Market equilibrium in power systems with network and storage constraints, **5)** Study case, **6)** Results, **7)** Conclusions and recommendations, **A.1)** Annex 1: Market Equilibrium Solutions.

1.1. Objectives

1.1.1. General

Formulate and solve the AC Optimal Power Flow problem for different market equilibrium points –Minimum Losses, Perfect Competition and Monopoly– considering three agents exposed to Locational Marginal Prices: Elastic demands, Renewable (Wind) producer without storage capabilities and Electric Vehicle aggregators.

1.1.2. Specific

- Discuss different market equilibrium points in traditional power systems without network and storage constraints.
- Apply the AC-OPF formulations in a illustrative 3-node power system.
- Perform a comparison analysis from the solutions obtained

2. Theoretical Framework

2.1. Background

Literature on market equilibrium models in electrical power system is vast. Interested reader can revisit [**ReferEnce**]. Focusing in models that incorporate storage at demand side, contributions as [He et al., 2015] show formulations in load shifting and loss reduction using energy storage devices. This contribution is meaningful since they investigate monopoly and oligopoly solutions for a network with batteries; however, the network constraints are ignored. Another contribution to the market equilibria analysis with storage devices is given by [Zhao et al., 2014]. They use direct revenue maximization for the battery owner using LMPs as price signals. LMPs must be a given value in order to maximize the revenue, and a series of market analysis are made that could lead to a 24 hours dispatch. In [Vani et al., 2012], it is proposed a heuristic optimization algorithm to find the LMPs of a complex system without direct calculation (losses incremental) nor auction-based simulation. Some recent contributions on OPF models presented in the University of Los Andes were also reviewed. A multiobjective OPF that aim to minimize system losses from the Distribution System Operator perspective was presented in [**TAPIA**]. A multiobjective OPF that aim to minimize system losses taking into account producers revenue was analyzed by [Toro, 2018]. A bilevel optimization approach for the coordination of a EV fleet in a 24-hour load curve was presented by [**TOQUICA**]. A comparative summary for some of the approaches to market equilibria and OPF can be found in table **2-1**.

Table 2-1.: Background in optimal power flow with network and storage constraints

Author	Optimization	Decision variables	Objective Function
S. Vani [**REF**]	BFOA	LMPs	Auction Replacement
Majid Baa [**REF**]	fmincon	P & Q	Voltage Stability
Hugo Tapia [**REF**]	Ant Colony	Q	Min Losses
Carlos Toro [**REF**]	Ant Colony	P & Q	Min Losses
David Toquica [**REF**]	Ant Colony	P	EV revenue Operational cost
Óscar Díaz	fmincon	LMPs, P & Q	Losses Producer Profit and Social Welfa

2.2. Optimization approach

Market equilibrium operating points are obtained by solving an optimization problem. For this purpose we employ the MATLAB fmincon's interior point algorithm. Here we provide details about the method used. According to the MATLAB toolbox support, the Optimization Toolbox provides functions for finding parameters that minimize or maximize objectives while satisfying constraints. The toolbox includes solvers for linear programming (LP), mixed-integer linear programming (MILP), quadratic programming (QP), nonlinear programming (NLP), constrained linear least squares, nonlinear least squares, and nonlinear equations. [MathWorks, 2019] This means that it has enough tools to solve a non-linear constraint problem, like the power flow with elastic demand that is proposed in this paper.

The function *fmincon* is one of the functions that the MATLAB Optimization Toolbox provides. It uses a series of numeric methods to solve a non-linear, bounded and constrained optimization. In this specific case, the default interior-point method is enough to solve the problem. This algorithm approach is to solve a sequence of approximate minimization problems using slack variables for each one of the constraints. To solve each one of the approximate problems, MATLAB solves either the KKT equations of a conjugate gradient problem, using a trust region. [MathWorks, 2019]

DocumentoGradoElectrica/images/Theoretical/fmincon.png

Figure 2-1.: MATLAB interior-point algorithm

Figure **2-1** depicts the fmincon path when solving a minimization for the MATLAB function *peaks()*. This algorithm will be used to solve the non-linear power flow and find the market equilibria that were proposed.

2.3. Optimal Power Flow for Minimal Losses

Taking the definition from [Glavitsch, 1991], the optimal power flow is an optimization problem that finds the value of some control variables that maximize or minimize an objective functions while taking into account the network physical constraints. This objective function can be related with system losses, voltage stability, generation cost, capacitor allocation, etc. One of the main applications of the OPF is the capacitor value calculation, this allows to find the reactive power in *var* that improves the most the desired voltage profile.

The following is an example of an OPF to minimize system losses, the constraints denoted as *System Model* are written in section 2.4.

$$\begin{aligned} \text{mín}\, P_L &= \frac{1}{2}\sum_{i=1}^{3}\sum_{k=1}^{3} G_{ij}[(V_i^t)^2 + (V_k^t)^2 - 2V_i^t V_k^t \cos(\theta_i^t - \theta_k^t)] \\ \text{s.t.} \quad & \text{The AC Network Model} \end{aligned}$$

This optimization will be studied further in section 6, since it will be the base scenario for the market analysis. For this case, the voltage and angles are constrained variables that will be determined from the power flow and the system active and reactive power are the decision variables.

2.4. The AC Network Model with Storage Constraints

For the sake of simplicity, we define a simple network model with three buses and three lines: Bus 1 is regarded as the grid supply point (slack bus) whose marginal cost corresponds to the wholesale electricity market. Bus 2 comprises an elastic demand and a wind producer without storage capabilities. Bus 3 comprises an elastic demand and a EV aggregator with storage capabilities. The network structure is given by the **admittance matrix**, in this case, a 3x3 complex matrix is defined in equation 2-2.

$$\mathbf{Y_{BUS}} = \mathbf{G} + \mathrm{j}\mathbf{B} \tag{2-1}$$

$$\mathbf{Y_{BUS}} = \begin{bmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \\ G_{31} & G_{32} & G_{33} \end{bmatrix} + \mathrm{j}\begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix} \tag{2-2}$$

Power flow constraints are shown from equations 2-3 through 2-8. These are an extended version of the classic power flow equations that include the time component. If we set $T = 24$, we will have the 1 day case of study. All the system voltages and angles are variables of the

optimization problem that will solve the power flow problem if we include those equations in the constraints.

$$
\begin{aligned}
P_1^t &= V_1^t \sum_{k\in\mathbf{N}} V_k^t[G_{1k}\cos\theta_{1k} + B_{1k}\sin\theta_{1k}] \quad \forall t \in \mathbf{T} && \text{(2-3)}\\
P_2^t - P_{D2}^{*,t} &= V_2^t \sum_{k\in\mathbf{N}} V_k^t[G_{2k}\cos\theta_{2k} + B_{2k}\sin\theta_{2k}] \quad \forall t \in \mathbf{T} && \text{(2-4)}\\
P_3^t - P_{D3}^{*,t} &= V_3^t \sum_{k\in\mathbf{N}} V_k^t[G_{3k}\cos\theta_{3k} + B_{3k}\sin\theta_{3k}] \quad \forall t \in \mathbf{T} && \text{(2-5)}\\
Q_1^t &= V_1^t \sum_{k\in\mathbf{N}} V_k^t[G_{1k}\sin\theta_{1k} - B_{1k}\cos\theta_{1k}] \quad \forall t \in \mathbf{T} && \text{(2-6)}\\
Q_2^t - Q_{D2}^{*,t} &= V_2^t \sum_{k\in\mathbf{N}} V_k^t[G_{2k}\sin\theta_{2k} - B_{2k}\cos\theta_{2k}] \quad \forall t \in \mathbf{T} && \text{(2-7)}\\
Q_3^t - Q_{D3}^{*,t} &= V_3^t \sum_{k\in\mathbf{N}} V_k^t[G_{3k}\sin\theta_{3k} - B_{3k}\cos\theta_{3k}] \quad \forall t \in \mathbf{T} && \text{(2-8)}\\
E_3^1 &= E_3^T && \text{(2-9)}\\
E_3^{t+1} &= E_3^t + P_3^t \quad \forall t \in \mathbf{T} && \text{(2-10)}
\end{aligned}
$$

Finally, equations 2-9 and 2-10 include the dynamic component of the problem, since they relate the battery state of charge and the power injected to node 3, as well as the states of charge over the T hours to optimize.

2.5. Locational Marginal Prices structure

The locational marginal prices (LMP) at given time t=1,...24 and node i=1,2.3 have three components: the system lambda, the incremental loss factor and the congestion fee:

$$\lambda_i^t = \lambda_0(1 - \eta_i^t) + \mu_i^t \tag{2-11}$$

In the following, we define each component:

2.5.1. The system lambda

For the purpose of this work, we consider the 24-hour marginal price profile obtained in the day ahead market as the system lambda.

2.5.2. Incremental losses

Usually, power systems calculate the Local Marginal Prices using the incremental value of the losses and the congestion penalty. To calculate the incremental losses value, a process

following the chain rule as in [Mutale et al., 2000] can be made. This calculation only works in small systems, since complex calculations have to be made if the Y_{bus} matrix is too complex. Such values can be found by the following relationship:

$$\begin{bmatrix} \eta_2^t \\ \eta_3^t \\ \gamma_2^t \\ \gamma_3^t \end{bmatrix} = \begin{bmatrix} \frac{\partial P_2^t}{\partial \theta_2^t} & \frac{\partial P_3^t}{\partial \theta_2^t} & \frac{\partial Q_2^t}{\partial \theta_2^t} & \frac{\partial Q_3^t}{\partial \theta_2^t} \\ \frac{\partial P_2^t}{\partial \theta_3^t} & \frac{\partial P_3^t}{\partial \theta_3^t} & \frac{\partial Q_2^t}{\partial \theta_3^t} & \frac{\partial Q_3^t}{\partial \theta_3^t} \\ \frac{\partial P_2^t}{\partial V_2^t} & \frac{\partial P_3^t}{\partial V_2^t} & \frac{\partial Q_2^t}{\partial V_2^t} & \frac{\partial Q_3^t}{\partial V_2^t} \\ \frac{\partial P_2^t}{\partial V_3^t} & \frac{\partial P_3^t}{\partial V_3^t} & \frac{\partial Q_2^t}{\partial V_3^t} & \frac{\partial Q_3^t}{\partial V_3^t} \end{bmatrix}^{-1} \begin{bmatrix} \frac{\partial L^t}{\partial \theta_2^t} \\ \frac{\partial L^t}{\partial \theta_3^t} \\ \frac{\partial L^t}{\partial V_2^t} \\ \frac{\partial L^t}{\partial V_3^t} \end{bmatrix} \tag{2-12}$$

Entries of the inverse of Jacobian and the right-hand vector are obtained from the partial derivatives using expressions:

$$\frac{\partial P_i^t}{\partial \theta_k^t} = V_i^t V_k^t [G_{ik} \sin(\theta_i^t - \theta_k^t) - B_{ik} \cos(\theta_i^t - \theta_k^t)] \tag{2-13}$$

$$\frac{\partial P_i^t}{\partial \theta_i^t} = -B_{ii}(V_i^t)^2 - \sum_{k=1}^{3} V_i^t V_k^t [G_{ik} \sin(\theta_i^t - \theta_k^t) - B_{ik} \cos(\theta_i^t - \theta_k^t)] \tag{2-14}$$

$$\frac{\partial P_i^t}{\partial V_k^t} = V_i^t [G_{ik} \cos(\theta_i^t - \theta_k^t) + B_{ik} \sin(\theta_i^t - \theta_k^t)] \tag{2-15}$$

$$\frac{\partial P_i^t}{\partial V_i^t} = G_{ii} V_i^t + \sum_{k=1}^{3} V_k^t [G_{ik} \cos(\theta_i^t - \theta_k^t) + B_{ik} \sin(\theta_i^t - \theta_k^t)] \tag{2-16}$$

$$\frac{\partial Q_i^t}{\partial \theta_k^t} = -V_i^t V_k^t [G_{ik} \cos(\theta_i^t - \theta_k^t) + B_{ik} \sin(\theta_i^t - \theta_k^t)] \tag{2-17}$$

$$\frac{\partial Q_i^t}{\partial \theta_i^t} = -G_{ii}(V_i^t)^2 + \sum_{k=1}^{3} V_i^t V_k^t [G_{ik} \cos(\theta_i^t - \theta_k^t) + B_{ik} \sin(\theta_i^t - \theta_k^t)] \tag{2-18}$$

$$\frac{\partial Q_i^t}{\partial V_k^t} = V_i^t [G_{ik} \sin(\theta_i^t - \theta_k^t) - B_{ik} \cos(\theta_i^t - \theta_k^t)] \tag{2-19}$$

$$\frac{\partial Q_i^t}{\partial V_i^t} = -B_{ii} V_i^t + \sum_{k=1}^{3} V_k^t [G_{ik} \sin(\theta_i^t - \theta_k^t) - B_{ik} \cos(\theta_i^t - \theta_k^t)] \tag{2-20}$$

$$\frac{\partial L^t}{\partial \theta_i^t} = 2 \sum_{k=1}^{3} V_i^t V_k^t G_{ik} \sin(\theta_i^t - \theta_k^t) \tag{2-21}$$

$$\frac{\partial L^t}{\partial V_i^t} = 2 \sum_{k=1}^{3} G_{ik} [V_i^t - V_k^t \cos(\theta_i^t - \theta_k^t)] \tag{2-22}$$

Incremental values will be used to calculate the prices in the base methodology (minimum losses), since for that base case the demand will be assumed as inelastic. Afterwards, the

demand will be assumed as elastic, which allows the Local Marginal Prices to be calculated from market equilibria between the demand and generation cost curves.

2.5.3. Congestion Fees

As said before, Local Marginal Prices can be decomposed in three main components: the system marginal price, the marginal loss component and the congestion fee. The last one is one of the main tools for the companies to take advantage from the network constraints, therefore it will be considered in the model. Taking the definition from [PJM, 2017], the congestion fee represents price of congestion for binding constraints and is calculated using the Shadow Price. Since it relies in the network constraints to be non-zero, it cannot be defined in a lossless model. Companies can use this fee strategically to create congestion in the grid and increase the Local Marginal Price of their node if they have enough power market to do it.

PLEASE INCLUDE HERE THAT LMPS CAN BE DETERMINED BY LAGRANGE MULTIPLIERS OR MODELING ALL DEMANDS IN THE OPF AS INELASTIC

2.6. Social Welfare concept

The introduction of smart meters and Advanced Measuring Infrastructure (AMI) in the demand side of the grid is allowing a demand response originated by price signals. According to [Barai et al., 2015], a smart meter is a device that includes sophisticated measurement and calculation hardware, software, calibration and communication capabilities. For interoperability within a smart grid infrastructure, smart meters are designed to perform functions, and store and communicate data according to certain standards.

Variation of power consumption on the demand side often include demand curtailment programs and price responsive demand programs. [Qinwei Duan, 2016] This means that the Independent System Operator, or other administrative roles, can have complete telemetry of the distribution network and send price signals to the consumers in real time in order to increase the reliability and efficiency of the power system. Advance Measuring Infrastructure and phasor Measurement Units can even allow the operator to build a complete Y_{bus} matrix of the system and run real-time simulations like the ones studied in this paper.

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Figure 2-2.: Elastic demand example

Based on the growing demand response programs, it makes sense to model the demand side load as an elastic demand curve. An example of decreasing curve that can model demand in an electric system is depicted in figure **2-2**. This curves follow the mathematical relationship stated below. Where λ_i is the Local Marginal Price, λ_i^{max} is the higher price that the demand is willing to pay (price for 0 units sold), m_d is the curve slope and x_d is the amount of units sold.

$$\lambda_i = \lambda_i^{max} - (x_d) \cdot m_d \tag{2-23}$$

PLEASE INCLUDE THE CONCEPT OF UTILITY OF DEMAND AND SOCIAL WELFARE

2.7. Energy Storage Model - 24 Hours Power Flow

To include the time-related effect of the energy storage device in the grid, it is necessary to run a dynamic power flow. This means that each variable will have **T** states per run and some relationship between states must be made. In this case, the battery *state of charge* will be related for **T=24** and a relationship with the power of the node will also be made. Equations 2-24 and 2-25 show a standard approach for this.

$$E_i^1 = E_i^T \tag{2-24}$$

$$E_i^{t+1} = E_i^t + P_i^t \quad \forall t \in \mathbf{T} \tag{2-25}$$

Dynamic power flow has a huge variety of applications, since it can be used together with OPF to optimize the system status as a whole and not hour by hour. Figure **2-3** describes an OPF that minimizes operational costs using MPC to improve the result of the system as a whole.

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Figure 2-3.: 24 hours power dispatch

3. Market equilibrium in power systems without network and storage constraints

Since the main goal of this thesis is to find market equilibrium solutions (Monopoly and Perfect Copetition) in power systems with network constraints, elastic demands and storage devices, it is important discuss before these concepts without network and storage constraints. In this chapter we define four equilibrium points (monopoly, Nash-Cournot, Stackelberg and Perfect Competition). Section **A.1** (Annex 1: Market Equilibrium Solutions) presents an analytic approach developed by [**De Oliveira-De Jesus, Paulo, "Notes for Course IELE4109 - Los Andes University", 2019**] for each of model assuming a simple two producer with an elastic demand without network and storage constraints.

3.1. Monopolistic Solution

Monopoly markets appear when a single company controls the production of a specific good. In the power market, this kind of behaviour can seriously harm the social welfare by rising the prices to maximize the producers welfare, specially in markets where there is not a demand response program. According to [Xin and Sainan, 2010], monopolies show the following characteristics in practice:

- Non-openness of the data, market information block of transactions and real estate transactions are carried out separately between developers and consumers.
- The Location Market Structure of the real estate oligopoly, makes monopoly profits exit in the real estate business, so the price of real estate transaction must exceed the value.
- There is a superiority of the seller bidding first in the real estate market, leading the real estate transaction price is higher than its equilibrium price

A complete development for this equilibrium using an analytic approach can be found in section A.1.3,

3.2. Nash-Cournot Equilibrium

A Cournot model is defined in [Xiong et al., 2009] as a game model which is about making decision of output. Nowadays, it is one of the classical models which research compete behavior of manufacturers in the duopolies market. In recent years, the Cournot model has been extended and has contained factors such as: incomplete information, dynamism and bounded rationality.

This kind of games can be solver by finding their Nash equilibrium. This equilibrium is a concept of game theory where the optimal outcome of a game is one where no player has an incentive to deviate from his chosen strategy after considering an opponent's choice. Given the complex matter of this approach, in this document we will only find this equilibrium in unconstrained models.

A complete development for this equilibrium using an analytic approach can be found in section A.1.5,

3.3. Stackelberg Equilibrium

The Stackelberg Equilibrium is a game by game theory approach. [Policonomics, 2019] define it as a sequential game (not simultaneous as in Cournot's model) where there are two firms, which sell homogeneous products, and are subject to the same demand and cost functions. One firm, the leader, is perhaps better known or has greater brand equity, and is, therefore, better placed to decide first which quantity to sell, and the other firm, the follower, observes this and decides on its production quantity

In [Lin et al., 2018] a complete analysis in oligopoly competition is found using Stackelberg theory. They explore the competitive mechanism of firms in a duopolistic market and constructs a model of duopolistic firms optimal output and maximum profit according to the Stackelberg Leadership Model. Given the complex matter of this approach, in this document we will only find this equilibrium in unconstrained models.

A complete development for this equilibrium using an analytic approach can be found in section A.1.6,

3.4. Perfect Competition

This type of equilibrium is the simplest and the most widely studied, as said in the paper of [Wang et al., 2018], the most important conditions for a perfect market to appear is that all

firms are price takers and every player has perfect information. Given that conditions and by the requirement that all firms have a relatively small market share, it is not possible to see a perfect competition in the real power market. Anyway, since this solution bring the highest of the social welfare among the equilibria, it can be used as a reference to calculate the deadweight loss of an imperfect market.

DocumentoGradoElectrica/images/Theoretical/PC-Industry-price-

Figure 3-1.: Example of perfect competition in a market

Finally, figure **3-1** shows the point of perfect competition in a supply vs demand graph. A complete development for this equilibrium using an analytic approach can be found in section A.1.2.

3.5. Comparison analysis

In section **A.1** (Annex 1: Market Equilibrium Solutions), an numerical example is provided for the above-defined equilibrium models. It is assumed a simple 1-hour market with two producers with an elastic demand without network and storage constraints. Figure **3-2** shows the results.

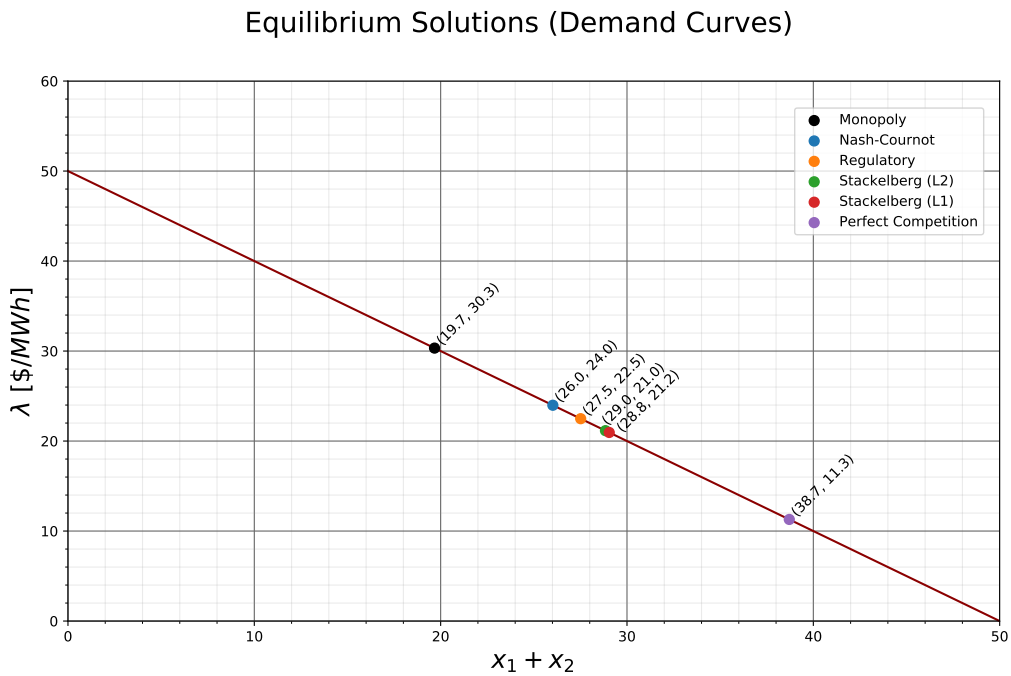

Figure 3-2.: Equilibrium solutions for a simple example

We can observe that perfect competition operating point corresponds to the best solution from social welfare viewpoint. Monopolistic solution is the best from producer viewpoint but a significant welfare loss since the demand is depressed and marginal prices the highest. Oligopolistic solutions are intermediate.

In the next chapter, we extend the analysis for constrained systems considering the extreme solutions depicted in Fig. **3-2**: Perfect competition and monopolistic solution. Oligopolistic solutions such as Nash and Stackelberg for constrained systems are out of scope of this work and matter of future research.

4. Market equilibrium in power systems with network and storage constraints

4.1. Network Model and Constraints

The formulation stated at 2.4 will be implemented in MATLAB in order to create a network model that can be solved using *fmincon*. Additional to the network constraints given by the Y_{bus} matrix and the power flow equations, the following will be included:

- **Simulation time:** Number of hours to simulate, leaving this as a parameter may help to generalize the study case in the future.
- **Minimum and maximum voltage:** Security constraints of voltage that every node in the system must fulfill. If the voltage in the nodes is too far away from the slack node, the system could become unstable and suffer a voltage collapse.
- **Battery capacity:** Maximum energy that the energy storage device can accumulate.
- **Battery maximum usage:** Percentage of the total battery capacity that the aggregator can use to buy and sell energy and make profit from it.
- **Minimum and maximum power:** Physical limitations of the batteries ripple and the wind production. A limit to the reactive power will also be set since the machines can not operate outside an specific power factor.
- **Line limits:** Since there are limits in the current that can flow through a grid line, it is necessary to include constraints in the maximum power that can flow. Changing those limits will allow us to include and analysis of the effect from line limits in the congestion fee.

4.2. Optimal Power Flow Problem

Each of the scenarios that we will study requires a different OPF formulation. In this section we will define the optimization problems assuming that the constraints denoted as *System Model* are the ones written in section 2.4.

4.2.1. Minimum Losses Dispatch

This scenario will be assumed as the base case, since it provide the Independent System Operator point of view for the economical dispatch. For this OPF an inelastic demand will be used and the prices will be calculated as the incremental value of the losses for each node, using the slack spot price. Based on the results from this OPF, the demand curve for the elastic cases will be calculated. Since the demand is assumed as inelastic using an arbitrary demand curve for the 24 hours, the social welfare result is not comparable with the other two cases. The mathematical formulation is shown below.

$$\text{mín}\, P_L = \frac{1}{2}\sum_{i=1}^{3}\sum_{k=1}^{3} G_{ij}[(V_i^t)^2 + (V_k^t)^2 - 2V_i^t V_k^t \cos(\theta_i^t - \theta_k^t)] \tag{4-1}$$

$$\text{s.t.} \tag{4-2}$$

$$V_i^{min} \leq V_i^t \leq V_i^{max} \quad \forall t \in \mathbf{T} \tag{4-3}$$

$$-Q_i^m \leq Q_i^t \leq Q_i^m \quad \forall t \in \mathbf{T} \tag{4-4}$$

$$-P_i^m \leq P_i^t \leq P_i^m \quad \forall t \in \mathbf{T} \tag{4-5}$$

$$\rho C_i \leq E_i^t \leq C_i \quad \forall t \in \mathbf{T} \tag{4-6}$$

$$S_{ij}^t \leq S_{ij}^{max} \tag{4-7}$$

ALL EQUATIONS SHOULD BE DISCUSSED IN DETAIL, ALL TERMS MUST BE DEFINED

Node 1 is the slack bus.

4.2.2. Maximum Social Welfare (Perfect Competition)

For this scenario, we will use the same model as the one in the minimum losses case, but assuming elastic demand. To calculate this demand curve, the average of power and Local Marginal Prices will be used and the slope will be left as parameter to explore different configurations. If the minimum losses scenario was to be tested with this elastic demand curve, the results for the power dispatch would be the same. The mathematical formulation is shown below.

$$\text{máx Social Welfare} = \text{Demand Utility - Producer costs} \tag{4-8}$$

$$\text{máx}\, SW = \sum_{i=2}^{N}[\lambda_i^{max} P_{di}^t - \frac{md_i {P_{di}^t}^2}{2}] - \sum_{i=2}^{N}[\lambda_i^{min} P_{gi}^t - \frac{mo_i {P_{gi}^t}^2}{2}] - \lambda_1^t P_1^t \tag{4-9}$$

$$\text{s.t.} \quad \text{The AC Network Model} \tag{4-10}$$

$$V_i^{min} \leq V_i^t \leq V_i^{max} \quad \forall t \in \mathbf{T} \tag{4-11}$$

$$-Q_{gi}^m \leq Q_{gi}^t \leq Q_{gi}^m \quad \forall t \in \mathbf{T} \tag{4-12}$$

$$-P_{gi}^m \leq P_{gi}^t \leq P_{gi}^m \quad \forall t \in \mathbf{T} \tag{4-13}$$

$$\rho C_i \leq E_i^t \leq C_i \quad \forall t \in \mathbf{T} \tag{4-14}$$

$$S_{ij}^t \leq S_{ij}^{max} \tag{4-15}$$

$$\lambda_i^t = \lambda_i^{max} - md_i \cdot P_{di}^t \tag{4-16}$$

ALL EQUATIONS SHOULD BE DISCUSSED IN DETAIL, ALL TERMS MUST BE DEFINED

Node 1 is the slack bus.

4.2.3. Monopoly Equilibrium

Finally, the monopoly equilibrium will be analyzed, the same model as the one in the maximum social welfare optimization will be used, but using a different objective function. This function is modeled as the sum of the generator profits for all the nodes that are owned by the monopoly company. In this equilibrium, the line flow constraints are even more important since the company can use them to increase the Local Marginal Price by the means of the congestion fee. The mathematical formulation is shown below.

$$\text{máx Total DG Producers Profit} = \text{DG Producer revenues - DG Producer Costs} \tag{4-17}$$

$$\text{máx}\, \Pi_{total} = \sum_{i=2}^{N}[\lambda_i^t P_{gi}^t] - \sum_{i=2}^{N}[\lambda_i^{min} P_{gi}^t - \frac{mo_i {P_{gi}^t}^2}{2}] \tag{4-18}$$

$$\text{s.t.} \quad \text{The AC Network Model} \tag{4-19}$$

$$V_i^{min} \leq V_i^t \leq V_i^{max} \quad \forall t \in \mathbf{T} \tag{4-20}$$

$$-Q_{gi}^m \leq Q_{gi}^t \leq Q_{gi}^m \quad \forall t \in \mathbf{T} \tag{4-21}$$

$$-P_{gi}^m \leq P_{gi}^t \leq P_{gi}^m \quad \forall t \in \mathbf{T} \tag{4-22}$$

$$\rho C_i \leq E_i^t \leq C_i \quad \forall t \in \mathbf{T} \tag{4-23}$$

$$S_{ij}^t \leq S_{ij}^{max} \tag{4-24}$$

$$\lambda_i^t = \lambda_i^{max} - md_i^t \cdot P_{di}^t \tag{4-25}$$

Node 1 is the slack bus.
ALL EQUATIONS SHOULD BE DISCUSSED IN DETAIL, ALL TERMS MUST BE DEFINED

4.3. Analysis and Comparison Procedure

For each scenario, the following variables will be calculated in a time horizon of 24 hours:

- System voltage in p.u. for each node
- Injected active and reactive power for each node
- Demand active and reactive power for each node
- Battery state of charge
- System line flows
- System Local Marginal Prices
- System hourly losses

In addition to this, the social welfare, total losses, total profit for each company, minimum, maximum and mean Local Marginal Prices will be calculated for each scenario. Finally, some variations in the system status, like disabling the energy storage device or including tight line constraints will be studied.

5. Study case

5.1. Three node example

In order to see the effects of the equilibria that are being studied, a 3 node example is defined including all the elements that we have considered up to this point. The following is a list of the characteristics for each of the nodes:

- **Node 1:** The spot market node, it represents the connection from the smart grid to the distribution network. It will be used as the slack node for the simulation and provides the base marginal price (λ_0).
- **Node 2:** The wind power farm. It has unidirectional active power flow for the generator and elastic demand.
- **Node 3:** The EV aggregator. Since it has a battery, the active power flow is bidirectional, it also has an elastic demand.

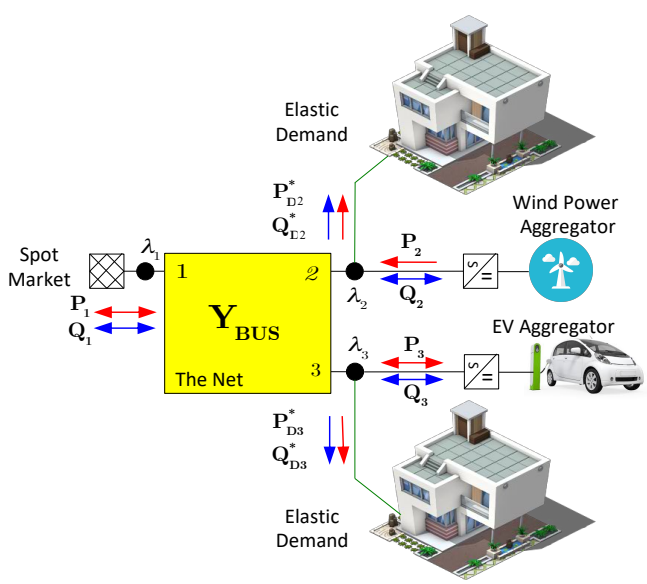

Figure 5-1.: System model used for the optimization

A graphic representation of the network is depicted in figure **5-1**.

5.1.1. Wind power curve

Since the wind farm has a fixed and predictable generation output based on the wind power, and does not have energy storage devices, the generated power for node 2 is determined by the curve in figure **5-2**.

DocumentoGradoElectrica/images/StudyCase/wind_farm_power.pdf

Figure 5-2.: Wind farm power curve

5.1.2. Elastic and inelastic demand

The study case comprises two types of demands. The minimum losses case assumes an inelastic demand curve that changes every hour regardless of the Local Marginal Price. On the other hand, the equilibrium cases assume elastic demand that changes as a function of the hourly LMP. Figure **5-3** shows both of the cases for nodes 2 and 3.

DocumentoGradoElectrica/images/StudyCase/Demand_curves.pdf

Figure 5-3.: Demand curves for elastic and inelastic case

5.1.3. Spot market price (λ_0)

Since the demand for the elastic case depends on the price, it is necessary to define the spot market price as a function of time. This means that every hour the base price that is used to calculate the LMP will change, therefore changing the demand. Figure **5-4** shows the curve that will be assumed for this study case.

DocumentoGradoElectrica/images/StudyCase/LAMBDA0_day.pdf

Figure 5-4.: Spot price curve for 24 hours

5.1.4. System matrix (Y_{BUS})

As stated before, the Y_{BUS} matrix defines the connections between the system nodes. It includes information about the real and imaginary part of the impedance for each one of the distribution lines, therefore defining the network constraints. Equation 5-1 shows the assumed structure for the network, implying that all the nodes are interconnected.

$$\mathbf{Y_{BUS}} = \begin{bmatrix} 53{,}33 & -13{,}3 & -40 \\ -13{,}3 & 4 & -26{,}67 \\ -4 & -26{,}6 & 66{,}67 \end{bmatrix} + \mathrm{j} \begin{bmatrix} -106{,}7 & 26{,}7 & 80 \\ 26{,}7 & -80 & 53{,}3 \\ 80 & 53{,}3 & -133{,}3 \end{bmatrix} \tag{5-1}$$

5.1.5. Operational limits

Finally, the network variables must be limited between some operational constraints in order to ensure the reliability and feasibility of the dispatch. This includes limits for the voltage, power and energy storage. Table **5-1** lists those limits for the study case.

Table 5-1.: Limits for the system variables

Variable	Minimum value	Maximum value
Nodal voltage	0.9 p.u.	1.1 p.u.
Battery capacity	80 MWh	80 MWh
Battery ρ_{BATT}	0.8	0.8
$\boldsymbol{Q_2}$	-10 Mvar	10 Mvar
$\boldsymbol{P_3}$	-10 MW	10 MW
$\boldsymbol{Q_3}$	-10 Mvar	10 Mvar

6. Results

6.1. Minimum losses scenario

The first scenario to be considered will be used as reference for the elastic demand curve as stated before. The 24 hour power flow was simulated and the results for the system voltages, active and reactive power, demand a battery state of charge is depicted in figure **6-1**.

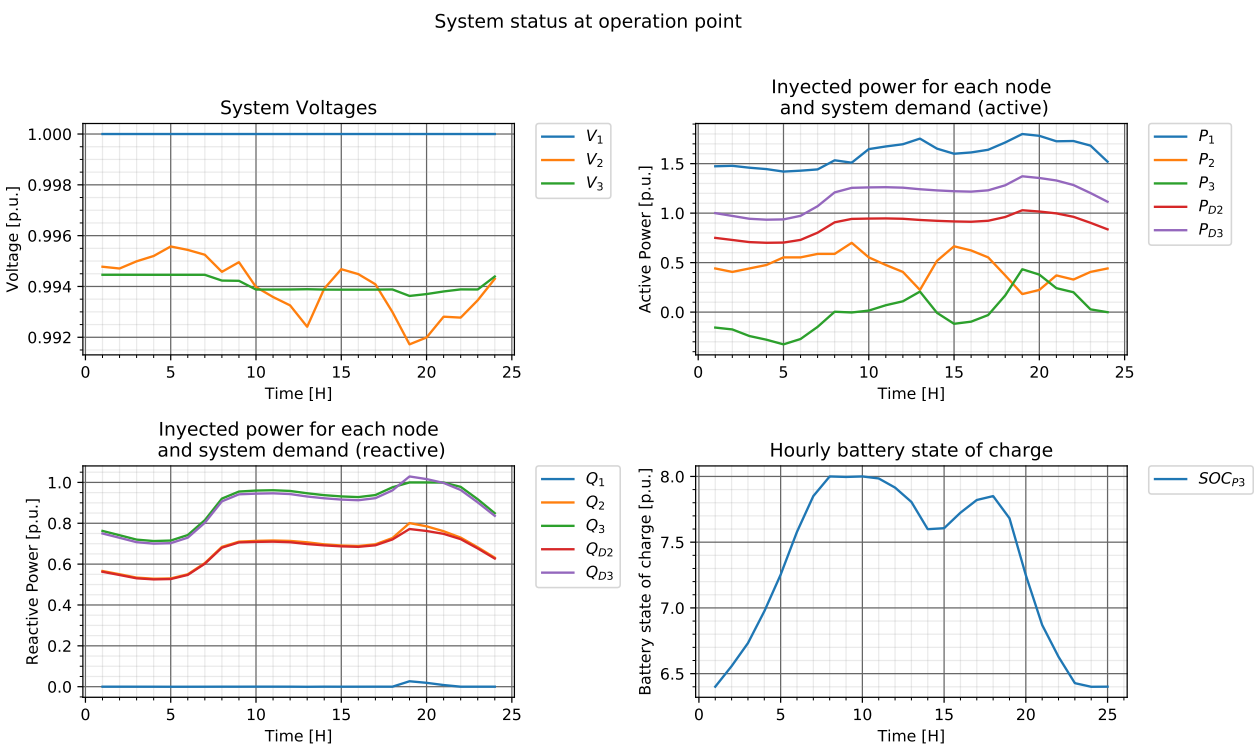

Figure 6-1.: System status: Minimum Losses

This results show that the system is highly compensated since this solution optimize the system status. The battery clearly tries to flatten the demand curve for node 3.

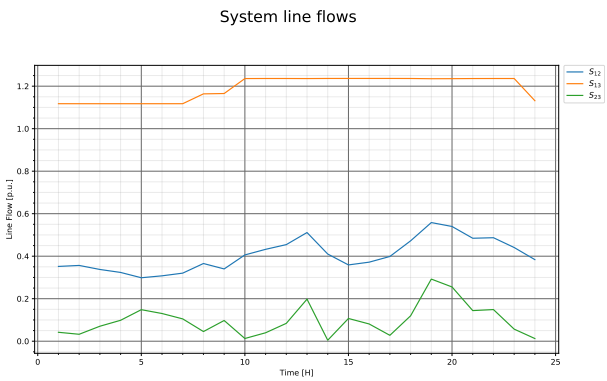

Figure 6-2.: System Line Flows: Minimum Losses

Figure 6-3.: Nodal Prices and System Losses: Minimum Losses

To completely determine the system status, we also calculated the system line flows that are shown in figure **6-2**. Nodal prices and system losses are important as well, since they are key to study the market equilibrium and compare it with the others, they are shown in figure **6-3**.

6.2. Perfect competition

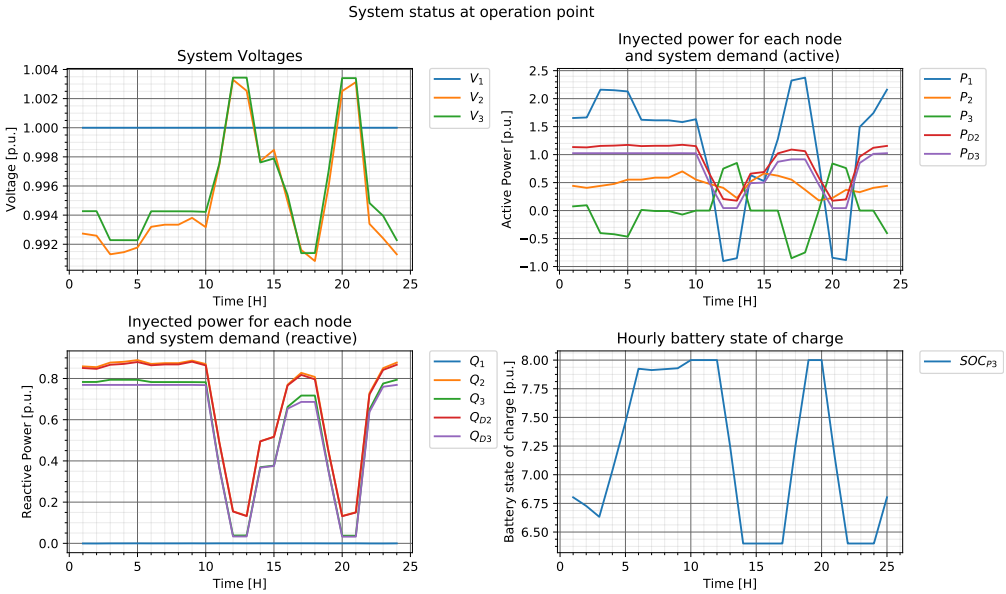

Figure 6-4.: System status: Maximum SW

Perfect competition is the second system status that we are studying, it follows the same principle as the minimum losses approach, but uses an economical approach. The power flow that was executed is depicted in figure **6-4**.

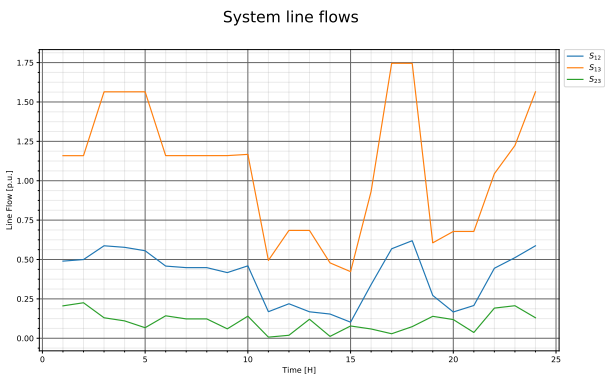

Figure 6-5.: System Line Flows: Maximum SW

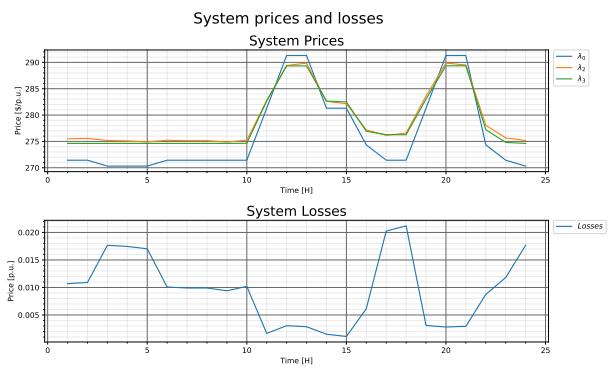

Figure 6-6.: Nodal Prices and System Losses: Maximum SW

Finally, the same calculations that the ones for the previous equilibrium are shown in figures **6-5** and **6-6**.

6.3. Monopoly scenario

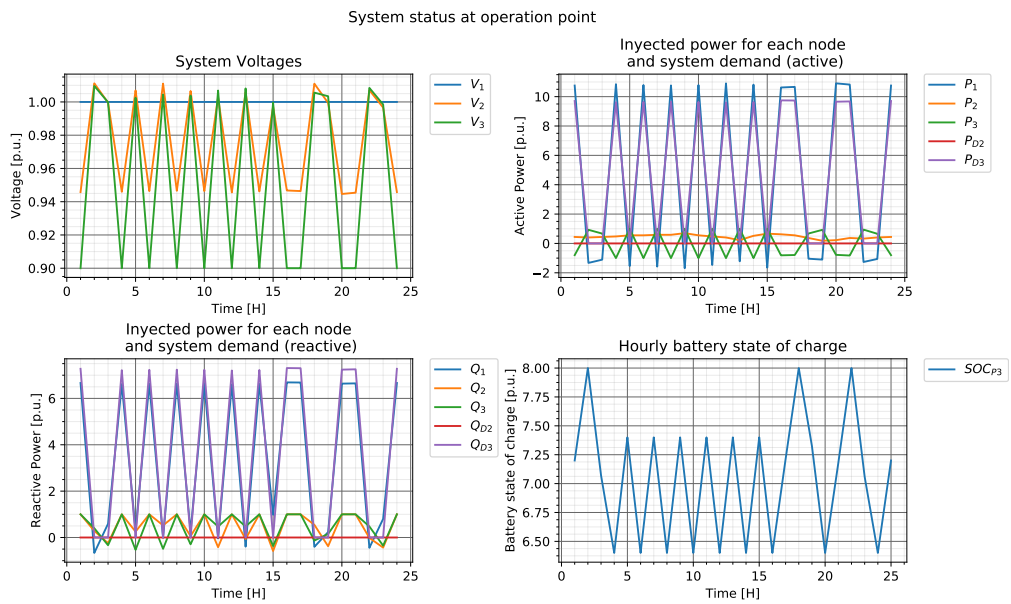

Figure 6-7.: System status: Maximum $\pi_2 + \pi_3$

The final equilibrium to be studied is the monopoly scenario. The erratic behaviour for all the system variables caused by the monopoly is shown in figure **6-7**.

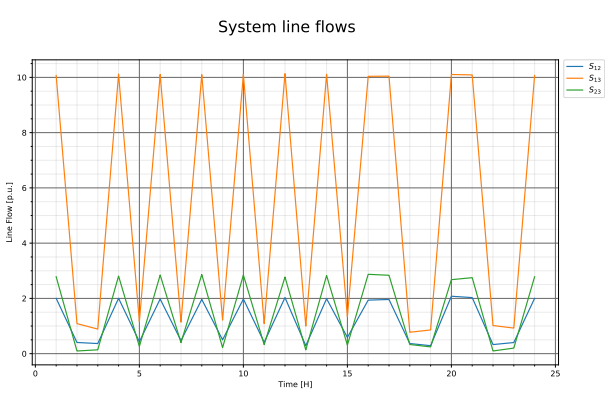

Figure 6-8.: System Line Flows: Maximum $\pi_2 + \pi_3$

Figure 6-9.: Nodal Prices and System Losses: Maximum $\pi_2 + \pi_3$

The same erratic behaviour can be seen in the system line flows and the Local Marginal Price. Figures **6-8** and **6-9** depict this.

6.4. Market comparison for the 3 scenarios

The minimum losses, maximum social welfare and monopoly power flows are shown as economic information in tables **6-1** and **6-2**.

Table 6-1.: Comparative table for the 3 optimization methods

Optimization	**SW**	**Losses**	π_2	π_3	$\pi_2 + \pi_3$
Min Losses (Inelastic)	-10,634.85	0.236	3,088.85	24.43	3,113.28
Max SW	3,483.78	0.228	3,085.80	44.46	3,130.26
Max $\pi_2 + \pi_3$	-6,243.21	8.469	3,243.24	565.66	808.90

Table **6-1** show the loss in social welfare produced by the monopoly solution, while increasing the profit for generators 2 and 3. Losses are considerably higher in the monopoly scenario.

That information is reaffirmed in table **6-2**, where the minimum, maximum and average Local Marginal Price for each node is shown.

Table 6-2.: Comparative table for the 3 optimization methods (LMP)

Optimization	λ_0			λ_2			λ_3		
	Min	**Max**	**Mean**	**Min**	**Max**	**Mean**	**Min**	**Max**	**Mean**
Min Losses (Inelastic)	270.31	291.31	276.44	272.76	296.12	279.79	273.37	294.97	279.86
Max SW	270.31	291.31	276.44	274.87	289.87	279.18	274.62	289.35	278.78
Max $\pi_2 + \pi_3$	270.31	291.31	276.44	292.50	292.50	292.50	143.82	290.00	217.49

6.5. Comparison of scenario variations

Some variations will be made on the results that were obtained to determine the effects that line flow limitations and lack of batteries cause in the result. Table **6-3** show the variations that were done in line flows for the system.

Table 6-3.: Scenarios for line limits

Limit	S_{12} (MVA)	S_{13} (MVA)	S_{23} (MVA)
Flow limit 1	5	1000	1000
Flow limit 2	1000	5	1000
Flow limit 3	5	5	5

6.5.1. Maximum social welfare

The first case of study in variations was the maximum social welfare scenario. Each of the entries in the table was compared with the base case for maximum social welfare through deadweight loss. Table **6-4** clearly shows the effect of each one of the variations.

Table 6-4.: Comparative table for scenario variations - SW

Variation	SW	Losses	π_2	π_3	$\pi_2 + \pi_3$	Deadweight loss
Max SW - Base	3,483.78	0.228	3,085.80	44.46	3,130.26	-
Max SW - No batteries	3,434.90	0.180	3,085.17	0.0	3,085.17	-48.88
Max SW - Flow limit 1	3,482.36	0.20	3,091.31	42.80	3,134.11	-1.42
Max SW - Flow limit 2	3,378.12	0.04	3,133.02	4.36	3,137.38	-105.68
Max SW - Flow limit 3	3,378.12	0.04	3,133.02	4.36	3,137.38	-105.66

Line limits affect both the social welfare and the profit for the generators since the total power that can be transmitted is less than optimal. The same table as before showing the Local Marginal Prices can be found in **6-5**.

Table 6-5.: Comparative table for scenario variations (LMP) - SW

Variation	λ_0			λ_2			λ_3		
	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
Max SW - Base	270.31	291.31	276.44	274.87	289.87	279.18	274.62	289.35	278.78
Max SW - No batteries	270.31	291.31	276.44	274.40	290.77	279.18	273.73	290.98	278.78
Max SW - Flow limit 1	270.31	291.31	276.44	274.99	289.86	279.67	274.83	289.35	278.95
Max SW - Flow limit 2	270.31	291.31	276.44	280.71	288.53	283.01	285.01	286.96	285.60
Max SW - Flow limit 3	270.31	291.31	276.44	280.71	288.53	283.01	285.01	286.96	285.60

6.5.2. Monopoly

Finally, the same comparison was made with the monopoly scenario. The losses in that scenario are clearly higher that the ones in perfect competition. Some increase in social welfare can be found in the line flow limits case since the total power that can be transmitted is less, diminishing the power market from the generators.

Table 6-6.: Comparative table for scenario variations - Monopoly

Variation	SW	Losses	π_2	π_3	$\pi_2 + \pi_3$	Deadweight loss
Monopoly - Base	-6,243.21	8.47	3,243.24	565.66	808.90	-9,726.99
Monopoly - No batteries	-9,531.41	11.21	3,243.23	0.0	3,243.23	-13,015.19
Monopoly - Flow limit 1	2,613.86	0.79	3,243.23	494.18	3,737.42	-869.92
Monopoly - Flow limit 2	3,120.90	0.11	3,241.18	30.93	3,272.11	-362.88
Monopoly - Flow limit 3	3,104.78	0.09	3,242.40	34.63	3,277.03	-379.00

Finally, the same table as before with the Local Marginal Prices can be found in **6-7**.

Table 6-7.: Comparative table for scenario variations (LMP) - Monopoly

Variation	$\boldsymbol{\lambda_0}$			$\boldsymbol{\lambda_2}$			$\boldsymbol{\lambda_3}$		
	Min	**Max**	**Mean**	**Min**	**Max**	**Mean**	**Min**	**Max**	**Mean**
Monopoly - Base	270.31	291.31	276.44	292.50	292.50	292.50	143.82	290.00	217.49
Monopoly - No batteries	270.31	291.31	276.44	292.50	292.50	292.50	163.73	174.79	166.62
Monopoly - Flow limit 1	270.31	291.31	276.44	292.50	292.50	292.50	241.41	290.00	268.74
Monopoly - Flow limit 2	270.31	291.31	276.44	281.20	292.49	292.02	278.45	289.99	283.45
Monopoly - Flow limit 3	270.31	291.31	276.44	291.29	292.50	292.44	279.72	289.99	284.95

7. Conclusions and recommendations

7.1. Conclusions

The main goal of analyzing and comparing the proposed market equilibria was achieved. A simulation and optimization tool was build using MATLAB and its toolbox and the results were validated by analyzing several variations of the base scenario.

The base optimization for the minimum losses scenario found a system set-point that settled the nodal voltage close to 1 p.u. and fed the demand of each node by using the local generation, looking forward to lower the flow through the system lines. This kind of actions also flattened the net demand of the nodes, which is a typical behaviour of the losses minimization problem. Since this case demand was assumed as inelastic, it was also a helpful tool to estimate the elastic demand curve. An important observation about the battery behaviour is that it was usually charging during the peak hours of the wind generation, but started to discharge when the wind was at its lower point.

The maximum social welfare scenario showed a similar behaviour in terms of the battery state of charge, but it was noticeable that both the power dispatch and the demand load were directly related with the spot market price. In terms of total losses, this dispatch should be exactly the same as the minimum losses dispatch if the system were exactly the same.

Finally, the monopoly scenario is an example of the damage that enough market power could do to a system. The voltages show a clear disruption from the nominal voltage, and the active and reactive power behaviour is highly erratic. The state of charge of the battery shows that it is completing as much battery cycles as possible, increasing the total profit for generator 3. This kind of behaviour together with abrupt changes in the Local Marginal Prices increases the total profit for the aggregator but could significantly lower the lifespan of the battery depending on the technology. Figure **6-9** also shows that the total system losses are considerably higher than in the social welfare scenario, same as the line flows through the system, which is an explanation for the increase in losses.

As for the Local Marginal Prices, table **6-2** shows that the mean price for both nodes 2 and 3 is higher than the ones in the maximum social welfare scenario, but the minimum price for the battery node is considerably lower, this is explained by the fact that the EV aggregator

prefers to charge the batteries at the lowest price possible and discharge them at the highest.

7.2. Recommendations and future work

Future implementations for this work should include a complete constrained model for the Nash-Cournot and Stackelberg equilibria. To accomplish that level of completeness, it may be necessary to include heuristic optimization to solve the bi-level problem that appears, some approaches like Ant Colony Optimization can be found in [Tapia, 2017].

Another improvement that could be made to this approach is to decouple the optimization problem from the power flow. This would increase the computational effort required but will also enable to write more complex optimization problems easily.

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A. Annexes

A.1. Annex 1: Market Equilibrium Solutions

In this section, a detailed mathematical solution will be given for each equilibrium. The example was taken form approach developed by [**De Oliveira-De Jesus, Paulo, "Notes for Course IELE4109 - Los Andes University", 2019**]. Each solution does not include network and storage constraints. The system that will be considered comprises 2 generation firms and 1 elastic demand in the same node. Every equilibrium will be presented as an optimization problem and a matrix solution will be given to calculate the generation of each firm and the nodal price for the system.

A.1.1. Producers and Demand Model

The costs structure for the generators in the grid is the traditional quadratic approach as shown below. The terms γ_i, α_i and β_i are used as known variables in all the equilibria. For the demand, a linear curve is used. Since the grid constraints are not considered, the total demand equals the total generation.

$$C_i(x_i) = \gamma_i + \alpha_i x_i + \frac{1}{2}\beta_i x_i^2$$

$$\lambda = \lambda^{max} - (x_1 + x_2) \cdot m_d \qquad\qquad x_d = x_1 + x_2$$

A.1.2. Perfect competition equilibrium

$$\begin{aligned} \text{máx} \quad & SW[x_1, x_2] = -C_1(x_1) - C_2(x_2) + \int \lambda^{max} - m_d x_d \quad dx_d \\ \text{subject to} \quad & x_1 + x_2 = x_d \end{aligned}$$

The perfect competition model implies that the *Social Welfare* must be maximum. Using the producers and demand model the optimization problem is set and the Lagrangian for the objective function is written:

$$L = \lambda^{max} x_d - \frac{1}{2} m_d x_d^2 - [\gamma_1 + \alpha_1 x_1 + \frac{1}{2}\beta_1 x_1^2 + \gamma_2 + \alpha_2 x_2 + \frac{1}{2}\beta_2 x_2^2] + \lambda[x_1 + x_2 - x_d]$$

Following the Lagrange method to solve non-linear optimization problems, the partial derivatives are calculated and equal to 0.

$$\begin{aligned}
\frac{\partial L}{\partial x_1} &= \lambda - \alpha_1 - \beta_1 x_1 = 0 \\
\frac{\partial L}{\partial x_2} &= \lambda - \alpha_2 - \beta_2 x_2 = 0 \\
\frac{\partial L}{\partial x_d} &= \lambda - \lambda^{max} + m_d x_d = 0 \\
\frac{\partial L}{\partial \lambda} &= x_d - x_1 - x_2 = 0
\end{aligned}$$

Finally, these equations are written as a matrix system to find a direct solution for the producers and demand set-points.

$$\begin{bmatrix} x_1^{pc} \\ x_2^{pc} \\ x_d^{pc} \\ \lambda^{pc} \end{bmatrix} = \begin{bmatrix} -\beta_1 & 0 & 0 & 1 \\ 0 & -\beta_2 & 0 & 1 \\ 0 & 0 & m_d & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \lambda^{max} \\ 0 \end{bmatrix}$$

A.1.3. Monopoly equilibrium

$$\begin{aligned}
\text{máx} \quad & \Pi[x_1, x_2] = \lambda x_1 - C_1(x_1) + \lambda x_2 - C_2(x_2) \\
\text{where} \quad & \lambda = \lambda^{max} - (x_1 + x_2) \cdot m_d
\end{aligned}$$

For this equilibrium, the addition of the producers profit must be at its maximum. Since the problem is unconstrained, a direct solution can be found by applying the first order conditions.

$$\begin{aligned}
\frac{\partial \Pi}{\partial x_1} &= \lambda^{max} - 2m_d x_1 - m_d x_2 - m_d x_2 - \alpha_1 - \beta_1 x_1 = 0 \\
\frac{\partial \Pi}{\partial x_2} &= \lambda^{max} - 2m_d x_2 - m_d x_1 - m_d x_1 - \alpha_2 - \beta_2 x_2 = 0
\end{aligned}$$

Finally, the matrix representation is presented to find the direct solution for the equilibrium.

$$\begin{bmatrix} x_1^m \\ x_2^m \end{bmatrix} = \begin{bmatrix} 2m_d + \beta_1 & 2m_d \\ 2m_d & 2m_d + \beta_2 \end{bmatrix}^{-1} \begin{bmatrix} \lambda^{max} - \alpha_1 \\ \lambda^{max} - \alpha_2 \end{bmatrix}$$

A.1.4. Regulatory Solution

Since the regulatory solution equals the average and marginal costs, no optimization is required and the problem can be set as a system of non-linear equations using the producer cost formulation.

$$
\begin{aligned}
AC_1 &= \frac{\gamma_1 + \alpha_1 x_1 + \frac{1}{2}\beta_1 x_1^2}{x_1} \\
AC_2 &= \frac{\gamma_2 + \alpha_2 x_2 + \frac{1}{2}\beta_2 x_2^2}{x_2} \\
AC_1 = AC_2 &= \lambda \\
\lambda &= \lambda^{max} - (x_1 + x_2) \cdot m_d
\end{aligned}
$$

This equilibrium can be found by solving the system of equations for any set of γ_i, α_i, β_i, m_d and λ^{max}.

A.1.5. Nash-Cournot equilibrium: Duopoly

$$
\begin{aligned}
\text{máx} \quad & \pi_i = \lambda x_i - C_i(x_i) \quad \forall i = \{1, 2\} \\
\text{subject to} \quad & \lambda = \lambda^{max} - (x_1 + x_2) \cdot m_d
\end{aligned}
$$

Since the Nash-Cournot equilibrium has to maximize the profits for producer 1 and 2 at the same time, both optimality conditions must be met. In order to use the first order conditions the derivatives for π_1 and π_2 were found.

$$
\frac{\partial \pi_i}{\partial x_i} = -m_d x_i + \lambda - \alpha_i - \beta_i x_i = 0
$$

$$
\begin{aligned}
\alpha_1 &= \lambda - (m_d + \beta_1) x_1 \\
\alpha_2 &= \lambda - (m_d + \beta_2) x_2 \\
\lambda^{max} &= \lambda + m_d x_1 + m_d x_2
\end{aligned}
$$

This system of equations can be represented as matrices to find a direct solution as follows.

$$
\begin{bmatrix} x_1^n \\ x_2^n \\ \lambda^n \end{bmatrix} = \begin{bmatrix} -m_d - \beta_1 & 0 & 1 \\ 0 & -m_d - \beta_2 & 1 \\ m_d & m_d & 1 \end{bmatrix}^{-1} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \lambda^{max} \end{bmatrix}
$$

A.1.6. Stackelberg equilibrium

In the Stackelberg formulation, the leader solution must be in the follower's solution set, therefore, the problem is presented as a bi-level optimization with *producer 2* as the leader and *producer 1* as the follower.

$$\begin{aligned} \text{máx} \quad & \pi_2 = x_2 \cdot [\lambda^{max} - m_d(x_1 + x_2)] - C_2(x_2) \\ \text{subject to} \quad & x_1 \equiv \text{argmax} \quad \pi_1 = x_1 \cdot [\lambda^{max} - m_d(x_1 + x_2)] - C_1(x_1) \end{aligned}$$

In order to solve the bi-level optimization using algebraic methods, it's necessary to find an expression for the follower solution in terms of the leader decision variable.

$$\begin{aligned} \frac{\partial \pi_1}{\partial x_1} &= \lambda^{max} - 2m_d x_1 - m_d x_2 - \alpha_1 - \beta_1 x_1 = 0 \\ x_1 &= A + Bx_2 = \frac{\lambda^{max} - \alpha_1}{2m_d + \beta_1} + \frac{-m_d}{2m_d + \beta_1} x_2 \end{aligned}$$

After finding that expression, it can be used to solve the univariate leader problem as follows.

$$\frac{\partial \pi_2}{\partial x_2} = \lambda^{max} - m_d A - \alpha_2 - [\beta_2 + 2m_d(B+1)] \cdot x_2 = 0$$

Finally, a direct solution for the problem is derived from this formulation.

$$\begin{aligned} x_2^{st} &= \frac{\lambda^{max} - m_d A - \alpha_2}{\beta_2 + 2m_d(B+1)} \\ x_1^{st} &= A + Bx_2^{st} \\ \lambda^{st} &= \lambda^{max} - m_d \cdot (x_1^{st} + x_2^{st}) \end{aligned}$$

A.1.7. Equilibrium Example: 2 Producers with Elastic Demand

In this section, an example will be given to present a numeric comparison of the equilibria solutions found above. The following values will be assumed for the producers and demand functions:

$$\begin{aligned} \text{Producer 1:} \quad & \gamma_1 = 0 \quad \alpha_1 = 10 \quad \beta_1 = 0{,}05 \\ \text{Producer 2:} \quad & \gamma_2 = 0 \quad \alpha_2 = 10 \quad \beta_2 = 0{,}1 \\ \text{Demand:} \quad & m_d = 1 \quad \lambda^{max} = 50 \end{aligned}$$

Since the regulatory solution sets the average cost equal to the marginal cost, this solution is completely dependent on γ_i and cannot be compared with the other results. For this example purposes, regulatory solution constants were assumed as: $\gamma_1 = 180$ *and* $\gamma_2 = 150$.

Each solutions yields to a different LMP which changes the *Social Welfare* in each case. Figure **A-1** depicts the position of the equilibria in the demand curve. As it can be seen, the monopoly equilibrium is in the leftmost part of the curve (less SW) and the perfect competition equilibrium is in the rightmost part of the curve (more SW). The Nash-Cournot and Stackelberg solutions are in the middle of those as expected.

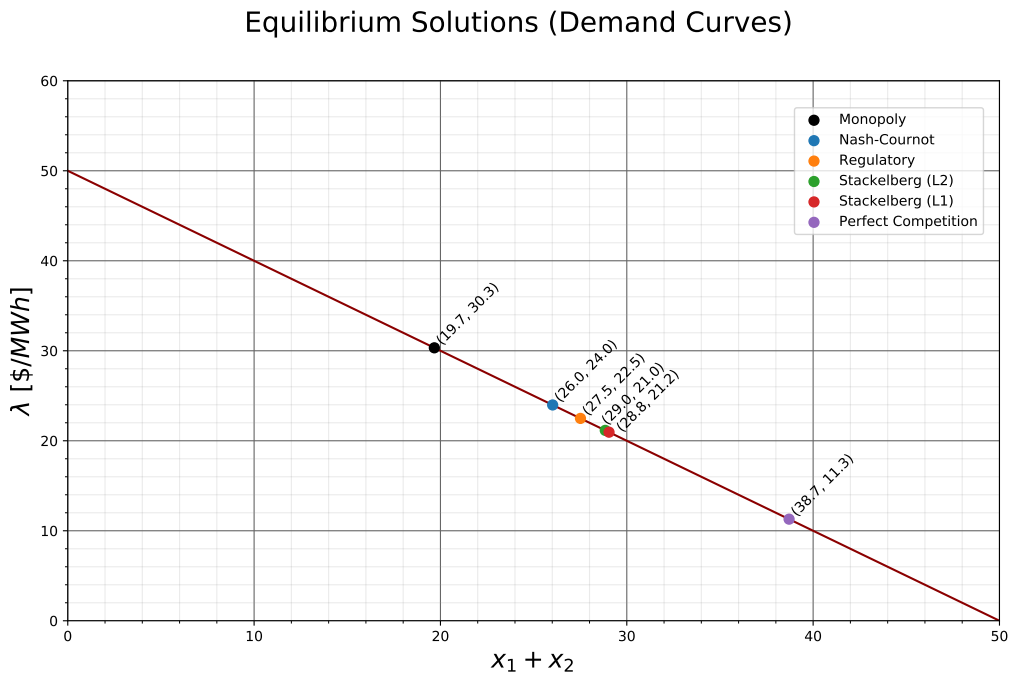

Figure A-1.: Equilibrium solutions for the example (demand)

More detailed comparison information is show in Tables **A-1** and **A-2**.

Table A-1.: Comparison between the studied equilibrium

Equilibrium	λ	π_1	π_2	π_{TOT}	x_1	x_2	x_d
Monopoly	30.33	262.30	131.15	393.44	13.11	6.56	19.67
Nash-Cournot	23.98	181.67	169.57	351.24	13.31	12.71	26.02
Regulatory	22.49	180.00	150.00	330.00	14.86	12.66	27.51
Stackelberg (P2)	21.15	115.69	186.66	302.34	10.62	18.22	28.85
Stackelberg (P1)	20.95	199.98	104.11	304.09	19.09	9.96	29.05
Perfect Competition	11.29	16.65	8.32	24.97	25.81	12.90	38.71

Table A-2.: Comparison between the studied equilibrium (producers and SW)

Equilibrium	**Benefit**	$\boldsymbol{CP_1}$	$\boldsymbol{CP_2}$	$\boldsymbol{CP_{TOT}}$	**SW**	**DL**
Monopoly	790.11	135.45	67.72	203.17	586.94	-187.25
Nash-Cournot	962.51	137.56	135.15	272.72	689.79	-84.40
Regulatory	997.21	154.11	134.56	288.68	708.53	-65.66
Stackelberg (P2)	1,026.23	109.06	198.81	307.87	718.36	-55.83
Stackelberg (P1)	1,030.48	200.00	104.53	304.53	725.94	-48.25
Perfect Competition	1,186.26	274.71	137.36	412.07	774.19	0.00